

GREIFSWALDER OIE, Germany

WMO: 100970

Lat: 54.2436N

Lon: 13.9103E

Elev: 11

StdP: 101.19

Time Zone: 1.00 (EUC)

Period: 01-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.8	-5.1	-10.4	1.6	-5.7	-8.5	1.8	-4.1	19.0	4.6	17.4	3.1	6.1	240	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.7	23.9	19.6	22.7	18.9	21.7	18.3	20.3	22.9	19.6	22.0	18.8	21.1	4.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.1	22.0	18.5	13.4	21.3	17.8	12.7	20.6	58.6	23.1	55.9	22.0	53.4	21.1	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	15.0	13.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-9.5	21.0	2.8	1.2	-11.5	21.9	-13.1	22.6	-14.7	23.3	-16.8	24.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	1.0	0.7	3.0	7.1	12.0	15.7	18.9	18.7	15.3	10.8	6.1	1.9		
	DBStd	7.09	3.45	2.86	2.85	2.58	2.55	2.10	2.31	1.95	2.15	2.70	2.44	3.46		
	HDD10.0	1251	277	260	217	93	11	0	0	0	0	24	118	251		
	HDD18.3	3366	536	493	475	338	196	82	21	19	93	235	368	509		
	CDD10.0	1003	0	0	0	5	73	172	274	271	159	47	1	0		
	CDD18.3	75	0	0	0	0	0	4	37	32	2	0	0	0		
	CDH23.3	62	0	0	0	0	1	1	48	13	0	0	0	0		
	CDH26.7	2	0	0	0	0	0	0	2	1	0	0	0	0		
(14) Wind	WSAvg	6.9	8.1	7.3	7.0	5.9	5.8	6.7	6.0	6.4	7.1	7.3	8.0	7.8		
(15) Precipitation	PrecAvg	601	47	36	38	33	50	61	68	65	57	47	48	50		
	PrecMax	762	91	55	85	103	114	144	246	168	127	82	113	117		
	PrecMin	453	4	6	10	4	15	0	0	23	14	16	6	12		
	PrecStd	83	23	12	20	20	23	28	49	33	28	18	21	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	8.6	10.7	15.5	19.7	23.0	26.2	24.6	21.9	17.5	12.3	10.1		
		MCWB	8.1	7.1	7.8	11.6	15.8	18.3	21.1	19.7	17.9	15.6	10.4	8.5		
	2%	DB	7.7	7.1	8.4	12.9	18.0	20.8	24.6	23.2	20.2	16.2	10.8	8.1		
		MCWB	6.3	5.9	6.6	10.1	14.8	17.3	20.1	19.0	16.9	14.4	9.7	6.9		
	5%	DB	6.7	6.0	7.4	11.5	16.5	19.7	23.4	22.3	19.4	15.3	10.0	6.9		
		MCWB	5.7	4.9	6.0	9.2	13.8	16.6	19.5	18.7	16.4	13.5	8.9	6.0		
	10%	DB	5.6	4.8	6.6	10.3	15.4	18.8	22.2	21.5	18.5	14.4	9.3	6.1		
		MCWB	4.7	3.9	5.3	8.3	13.0	15.9	18.7	18.4	15.9	12.7	8.3	5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	7.3	8.2	11.8	16.7	19.1	21.8	20.7	18.6	16.3	10.7	8.5		
		MCDB	9.8	8.2	9.7	14.8	18.8	21.6	25.3	23.0	21.0	17.1	11.6	9.4		
	2%	WB	6.6	6.1	7.0	10.5	15.3	17.9	20.9	20.0	17.7	14.6	9.9	7.1		
		MCDB	7.5	6.9	8.0	12.7	17.3	20.0	23.7	22.1	19.6	15.9	10.8	8.0		
	5%	WB	5.7	5.0	6.1	9.5	14.3	17.1	20.0	19.5	17.0	13.6	9.2	6.1		
		MCDB	6.6	5.9	7.3	11.2	16.1	19.4	23.1	21.5	18.9	15.0	10.0	6.8		
	10%	WB	4.7	4.0	5.4	8.6	13.2	16.3	19.1	18.9	16.3	12.9	8.4	5.4		
		MCDB	5.5	4.8	6.4	10.1	15.2	18.5	21.6	21.0	18.3	14.3	9.2	6.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.6	2.5	3.2	3.8	4.1	4.1	4.1	3.7	3.5	3.0	2.5	2.6		
		MCDBR	3.1	3.2	4.6	5.4	5.8	5.2	5.2	4.6	4.4	3.7	2.6	2.7		
	5% WB	MCWBR	2.7	2.8	3.4	3.3	3.8	3.3	3.1	3.0	2.9	2.7	2.5	2.4		
		MCWBR	3.0	3.0	4.5	5.0	5.3	4.9	4.8	4.1	4.0	3.5	2.5	2.6		
(40) Clear-Sky Solar Irradiance	taub		0.319	0.342	0.371	0.402	0.399	0.395	0.416	0.415	0.381	0.364	0.342	0.313		
		taud	2.329	2.331	2.285	2.270	2.337	2.378	2.332	2.343	2.414	2.405	2.365	2.320		
	Ebn at Noon		628	740	798	823	847	855	826	803	785	707	593	551		
		Edh at Noon	57	79	102	118	117	113	117	109	90	73	56	48		
(44) All-Sky Solar Radiation	RadAvg		0.50	1.13	2.40	4.17	5.34	5.69	5.24	4.30	3.10	1.58	0.64	0.35		
	RadStd		0.06	0.19	0.22	0.49	0.59	0.41	0.62	0.33	0.30	0.23	0.08	0.04		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (50 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-156	+58	N/A	

Nomenclature: See separate page